Analytic Stress Tensor in DFT+eDMFT with LAPW

Stationary-functional derivation, precise equations, and a source-level implementation plan for `dmft2`

Revision 2: fixed LAPW geometry conventions and local-axis treatment

Based on the DFT+eDMFT force formalism and the WIEN2k LAPW stress implementation

Purpose. This note identifies the stress equations that follow from the stationary DFT+eDMFT Klein functional, separates the terms already handled by `lapw0`, `lcore`, and the mixer, maps the reusable WIEN2k `lapw2` routines, inventories the corresponding machinery already present in `dmft2`, and gives a staged implementation and validation plan. It is a design document only; no source-code changes are made.

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1 Executive summary

The most economical and safest implementation is not a second, independent stress code. It is a generalization of the existing WIEN2k valence-stress path inside `dmft2`, combined with the existing `lapw0` and `lcore` stress components.

The final stress should be organized as

$$\boxed{\sigma_{\alpha\beta}^{\mathrm{DFT+eDMFT}} = \sigma_{\alpha\beta}^{\mathrm{val,kin}}[\rho] + \sigma_{\alpha\beta}^{\mathrm{val,corr}}[\rho, \rho^{E}] + \sigma_{\alpha\beta}^{\mathrm{proj}} + \sigma_{\alpha\beta}^{\mathrm{core,corr}} + \sigma_{\alpha\beta}^{\mathrm{es}} + \sigma_{\alpha\beta}^{\mathrm{xc}}.} \tag{1}$$

The ownership of these pieces should be:

Contribution	Correct implementation	Status
Valence kinetic	generalized WIEN2k `lapw2` kernels, contracted with the interacting density matrix ρ	add to `dmft2`
Valence IBS/Pulay correction	generalized `val_corr_stress`, with separate H and overlap kernels contracted with ρ and ρ^{E}	add to `dmft2`
Explicit DMFT projector stress	derivative of the normalized finite-window projector in the embedded self-energy	new DMFT term
Electrostatic and XC	existing `lapw0` stress evaluated with the converged eDMFT density	already available
Core correction	existing `lcore` nonspherical core-stress correction	already available
Final assembly and sign convention	existing mixer: valence + core − `STR_DENS`	already available

Central implementation result. The ordinary DFT occupation $f_{n\mathbf{k}}$ must not simply be replaced by a "DMFT occupation." Two different matrices are required: the interacting density matrix $\rho_{\mathbf{k}}$ and the energy-weighted matrix $\rho_{\mathbf{k}}^{E}$. The first multiplies Hamiltonian-like and kinetic kernels; the second multiplies overlap/Pulay kernels. Both matrices already exist in `dmft2` as `Aweight` and `AEweight`.

Already present in `dmft2`. Interacting density and energy-density matrices, arbitrary Hermitian pseudo-band transformations, off-diagonal force contractions, normalized projectors, their force-normalization derivative, and a finite-window response using the extra bands stored in `case.vector` or `case.vectorso`.

Main missing work. Import the WIEN2k stress kernels and their radial/plane-wave preparation; separate the valence correction into H and overlap parts; add six strain derivative channels for the projector, including fixed-domain geometry, finite-window response, and normalization; generalize the hard-coded force-channel/logarithm machinery; reduce, symmetrize, and write `:STR_VAL001-003`.

Conventions fixed in this revision

The first implementation will use the following precise derivative path:

Object	Convention for the analytic derivative
Muffin-tin radius	R_{MT} is fixed in bohr; $\partial_q R_{\mathrm{MT}} = 0$.
Muffin-tin domain	current spherical domain centered on the affinely displaced atom, not a material sphere pulled into an ellipsoid
Projectors 5/6	the radial profile, `projectorw.dat`, R_{MT}, and the outer radius R_2 are fixed; only their embedding in the strained LAPW basis is differentiated
Band window	fixed band-index interval; no energy-cutoff derivative, but mixing with stored bands outside the DMFT interval is included
Local correlated axes, version 1	fixed in global Cartesian space, so their explicit derivative is zero
Local correlated axes, later extension	optional geometry-following frame, needed when a diagonal-only self-energy is intended to remain diagonal in an instantaneous ligand/crystal-field frame

Important consequence of the diagonal self-energy approximation. With a full rotationally covariant local self-energy, a rotation of the complete correlated shell is a gauge choice. With a self-energy constrained to be diagonal, the local axes define the approximation and are therefore physical input. Version 1 will differentiate the well-defined approximation with fixed Cartesian axes. A geometry-following-axis derivative is a separate, explicit projector term and can be added after the base stress is validated.

2 Scope, conventions, and the sign of stress

We use the same convention as the Blaha stress derivation:

$$\sigma_{\alpha\beta} = \frac{1}{\Omega} \left. \frac{\partial \Gamma[\epsilon]}{\partial \epsilon_{\alpha\beta}} \right|_{\epsilon=0}, \tag{2}$$

where Γ is the stationary thermodynamic functional. In DFT it reduces to the usual total-energy or finite-temperature free-energy functional; in DFT+eDMFT it is the stationary Klein functional. With this convention,

$$P = -\frac{1}{3} \operatorname{Tr} \sigma. \tag{3}$$

The infinitesimal strain is taken to be symmetric. For compact storage we use

$$q = 1, \ldots, 6 \quad \longleftrightarrow \quad (xx, yy, zz, yz, xz, xy). \tag{4}$$

For an off-diagonal component the derivative is the derivative with respect to the symmetric strain variable. This formula follows directly from the deformation gradient

$$F(\epsilon) = I + \epsilon, \qquad \mathbf{r}^{\epsilon} = F\mathbf{r}, \qquad \mathbf{G}^{\epsilon} = F^{-T}\mathbf{G}. \tag{5}$$

For a symmetric independent variable,

$$\frac{\partial \epsilon_{\gamma\delta}}{\partial \epsilon_{\alpha\beta}} = \frac{1}{2}\left(\delta_{\gamma\alpha}\delta_{\delta\beta} + \delta_{\gamma\beta}\delta_{\delta\alpha}\right). \tag{6}$$

Applying Eq. (6) to Eq. (5) gives

$$\partial_{\alpha\beta} V_\gamma = \pm\frac{1}{2}\left(\delta_{\gamma\alpha}V_\beta + \delta_{\gamma\beta}V_\alpha\right), \tag{7}$$

with the plus sign for a real-space vector and the minus sign for a reciprocal vector. This is Appendix A, Eq. (A1), of the Blaha derivation; it is not an additional physical assumption. For example, $\partial_{xy}V_x = \pm V_y/2$ and $\partial_{xy}V_y = \pm V_x/2$. Also

$$\partial_{\alpha\beta}\Omega = \delta_{\alpha\beta}\Omega. \tag{8}$$

Shear-factor trap. Finite differences must use exactly the convention in Eq. (7). A finite difference that directly returns σ_{xy} should use $E^{xy} = (\mathbf{e}_x \otimes \mathbf{e}_y + \mathbf{e}_y \otimes \mathbf{e}_x)/2$. If instead one sets $F_{xy} = F_{yx} = s$, then $\mathrm{d}E/\mathrm{d}s = 2\Omega\sigma_{xy}$. Applying both entries as independent full changes and comparing with one tensor component produces a factor-of-two error.

WIEN2k writes full 3×3 tensors. Internally, however, only six symmetric derivatives are independent. The implementation should store six derivative channels and expand them into a symmetric 3×3 tensor only for symmetrization and output.

3 The precise LAPW geometry differentiated in version 1

An analytic stress is the derivative along a specified numerical path. The LAPW partition, projector support, and local frame must therefore be defined before differentiating them. The following choices reproduce how a strained WIEN2k calculation is actually rebuilt and avoid introducing derivatives of arbitrary basis parameters.

3.1 Fixed muffin-tin radii

For every atom a,

$$R_a^\epsilon = R_a, \qquad \partial_q R_a = 0. \tag{9}$$

R_a is a numerical partition parameter, not a material coordinate. Scaling it with the cell would define a different basis path and would require extra radial and surface derivatives. Therefore every analytic/finite-difference comparison must use the same radii for the $-\delta$, 0, and $+\delta$ structures. If the strained spheres overlap, choose a smaller radius and use it for all three calculations rather than changing it only at one strain.

3.2 Current spherical muffin-tin domains

The center moves affinely,

$$\boldsymbol{\tau}_a^\epsilon = F\boldsymbol{\tau}_a, \tag{10}$$

but the integration domain is the current sphere

$$\mathcal{D}_a^\epsilon = \{\mathbf{r}^\epsilon : |\mathbf{r}^\epsilon - \boldsymbol{\tau}_a^\epsilon| < R_a\}. \tag{11}$$

Introduce the current atom-centered coordinate $\mathbf{x} = \mathbf{r}^\epsilon - \boldsymbol{\tau}_a^\epsilon$. Then

$$\int_{\mathcal{D}_a^\epsilon} \mathrm{d}^3 r^\epsilon \, f^\epsilon(\mathbf{r}^\epsilon) = \int_{|\mathbf{x}|<R_a} \mathrm{d}^3 x \, f^\epsilon(\boldsymbol{\tau}_a^\epsilon + \mathbf{x}). \tag{12}$$

The domain on the right is independent of strain. Consequently, a projector that is finite at R_a does *not*, by that fact alone, create a Reynolds surface term in this convention. Its derivative is obtained by differentiating the integrand at fixed $\mathbf{x}$. A material-domain pullback, $F\mathcal{D}_a^0$, would instead make the sphere an ellipsoid under shear and would distribute equivalent terms between volume and surface pieces. It is not the representation used for version 1.

Surface terms are representation dependent. When a whole-cell integral is split into muffin-tin and interstitial pieces, interface terms may appear and must cancel or be retained exactly as in the validated WIEN2k kernel. The statement above means only that one must not append a new projector boundary term merely because the radial projector is nonzero at R_{MT}.

3.3 Fixed radial projectors, including projectors 5 and 6

For the initial implementation,

$$\phi_{alm}^\epsilon(\mathbf{x}) = \phi_{al}(|\mathbf{x}|) Y_{lm}(\hat{\mathbf{x}}), \qquad \partial_q \phi_{alm}^\epsilon(\mathbf{x})\big|_{\mathbf{x}} = 0, \tag{13}$$

apart from an explicitly chosen local-axis rotation discussed later. The same `projectorw.dat`, radial mesh, R_{MT}, and outer radius R_2 are used at all strains. In the present projector reader the inner mesh ends at R_{MT} and the tail mesh spacing is constructed from $R_2 - R_{\mathrm{MT}}$; holding the file and both radii fixed therefore defines the derivative unambiguously.

When a projector 5/6 tail extends beyond R_{MT}, the muffin-tin boundary is an interface inside the support, not a cutoff boundary. The inside and tail representations must be differentiated together. The direct strain derivative then acts on the LAPW/plane-wave embedding—reciprocal vectors, Bessel and angular factors, phases, normalization, and matching—but not on a newly regenerated radial solution. Regenerating `projectorw.dat` at every strain would define a potential-dependent projector and would require a separate radial-response calculation.

3.4 Fixed local axes for the base implementation

Let R_a^{loc} denote the correlated local frame read from the structure/DMFT input. Version 1 uses

$$R_a^{\mathrm{loc}}(\epsilon) = R_a^{\mathrm{loc}}(0), \qquad \partial_q R_a^{\mathrm{loc}} = 0, \tag{14}$$

where the matrix is understood in global Cartesian coordinates. Symmetric strain contains no rigid-body rotation. Thus this choice is exact for hydrostatic strain and natural for axis-aligned tetragonal tests; it is also a clean, reproducible definition for shear. A geometry-following axis is a later extension, not an ambiguity in version 1.

4 Why the stationary Klein functional must be differentiated

The eDMFT total-energy expression is not stationary with respect to every intermediate numerical variable. Differentiating it directly would introduce uncontrolled derivatives of G, Σ, the impurity solution, and the double-counting potential. The force derivation already solved this problem by starting from the stationary DFT+eDMFT Klein functional.

A schematic form sufficient for the present derivation is

$$\begin{aligned}\Gamma[G] = & -\operatorname{Tr}\ln[-G^{-1}] - \operatorname{Tr}[(G_0^{-1} - G^{-1})G] + E_H[\rho] + E_{\mathrm{xc}}[\rho] + E_{\mathrm{ion}} \\ & + \Phi_{\mathrm{DMFT}}[G_{\mathrm{loc}}] - \Phi_{\mathrm{DC}}[n_{\mathrm{loc}}].\end{aligned} \tag{15}$$

At self-consistency,

$$\frac{\delta\Gamma}{\delta G} = 0, \qquad \frac{\delta\Phi_{\mathrm{DMFT}}}{\delta G_{\mathrm{loc}}} = \Sigma, \qquad \frac{\partial\Phi_{\mathrm{DC}}}{\partial n_{\mathrm{loc}}} = V_{\mathrm{DC}}. \tag{16}$$

Therefore, for any strain component q,

$$\frac{\mathrm{d}\Gamma}{\mathrm{d}\epsilon_q} = \left.\frac{\partial\Gamma}{\partial\epsilon_q}\right|_{G,\Sigma,n_{\mathrm{loc}}} + \underbrace{\frac{\delta\Gamma}{\delta G}\frac{\mathrm{d}G}{\mathrm{d}\epsilon_q}}_{0} + \underbrace{\left(\frac{\delta\Phi}{\delta G_{\mathrm{loc}}} - \Sigma\right)\frac{\mathrm{d}G_{\mathrm{loc}}}{\mathrm{d}\epsilon_q}}_{0}. \tag{17}$$

This is the stress analogue of the force cancellation. The impurity two-particle vertex is not needed for a first derivative. It would enter an analytic elastic constant, which is a second derivative of the stationary functional.

Consequence. For stress we differentiate only the explicit geometric dependence of the LAPW basis, volume, reciprocal vectors, muffin-tin partition, and normalized projector. We do not differentiate the converged self-energy by solving an impurity vertex equation.

This stationarity statement also applies to a diagonal-only self-energy provided that the approximation is defined by a restricted local functional, for example a functional of the diagonal projected Green functions in a specified local frame. Then the embedded diagonal self-energy is the derivative of that restricted functional and no vertex is required for the first strain derivative. If off-diagonal self-energy components are instead discarded as an ad hoc post-processing operation that is not the derivative of any approximate functional, exact stationarity is lost; the residual $\delta\Gamma/\delta G$ would then multiply $G_{;q}$. The implementation should therefore document that its diagonal/block structure is a definition of the approximate functional, not merely a numerical truncation after the impurity solution.

5 The two eDMFT one-particle matrices

At each $\mathbf{k}$ point, define the interacting band-space density matrix

$$\rho_{\mathbf{k}} = \frac{1}{\beta}\sum_n e^{i\omega_n 0^+} G_{\mathbf{k}}(i\omega_n), \tag{18}$$

and the corresponding first energy moment

$$\rho^E_{\mathbf{k}} = \frac{1}{\beta}\sum_n e^{i\omega_n 0^+}\left[H^{\mathrm{KS}}_{\mathbf{k}} + \Sigma^{\mathrm{emb}}_{\mathbf{k}}(i\omega_n)\right] G_{\mathbf{k}}(i\omega_n). \tag{19}$$

Equation (19) is a compact formal definition. In code it is better to use the already regularized high-frequency expression that constructs `AEweight`; it has the correct chemical-potential and Matsubara-tail convention.

In the DFT limit,

$$(\rho_{\mathbf{k}})_{ij} = w_{\mathbf{k}} f_{i\mathbf{k}}\delta_{ij}, \qquad (\rho^E_{\mathbf{k}})_{ij} = w_{\mathbf{k}} f_{i\mathbf{k}}\varepsilon_{i\mathbf{k}}\delta_{ij}. \tag{20}$$

The basic nonorthogonal-basis derivative then has the universal form

$$\boxed{\Omega\,\sigma^{\mathrm{one}}_q = \sum_{\mathbf{k}} \mathrm{Tr}\left[\rho_{\mathbf{k}} H_{\mathbf{k};q} - \rho^E_{\mathbf{k}} O_{\mathbf{k};q}\right],} \tag{21}$$

where $H_{;q}$ and $O_{;q}$ mean explicit geometric derivatives of the LAPW Hamiltonian and overlap. Equation (21) is the equation that must replace the DFT band sum $\sum_n f_n(H_{nn;q} - \varepsilon_n O_{nn;q})$.

In the present `dmft2` source:

$$\rho_{\mathbf{k}} \longleftrightarrow \texttt{Aweight}, \qquad \rho^E_{\mathbf{k}} \longleftrightarrow \texttt{AEweight}. \tag{22}$$

Both include the **k**-point weight in the current implementation. A new stress path must preserve that convention and must not multiply by the weight again.

6 The DFT LAPW stress decomposition to preserve

The Blaha derivation gives

$$\sigma_{\alpha\beta} = \sigma^{\mathrm{val,kin}}_{\alpha\beta} + \sigma^{\mathrm{val,corr}}_{\alpha\beta} + \sigma^{\mathrm{core,corr}}_{\alpha\beta} + \sigma^{\mathrm{es}}_{\alpha\beta} + \sigma^{\mathrm{xc}}_{\alpha\beta}. \tag{23}$$

For DFT occupations $f_{n\mathbf{k}}$, the first three explicit formulas are

$$\sigma^{\mathrm{val,kin}}_{\alpha\beta} = \frac{1}{2\Omega}\sum_{n\mathbf{k}} f_{n\mathbf{k}}\langle\psi_{n\mathbf{k}}|\partial_\alpha\partial_\beta + \partial_\beta\partial_\alpha|\psi_{n\mathbf{k}}\rangle, \tag{24}$$

$$\sigma^{\mathrm{val,corr}}_{\alpha\beta} = \frac{2}{\Omega}\sum_{n\mathbf{k}} f_{n\mathbf{k}}\,\mathrm{Re}\left[\langle\partial_{\alpha\beta}\psi_{n\mathbf{k}}|H_{\mathrm{eff}} - \varepsilon_{n\mathbf{k}}|\psi_{n\mathbf{k}}\rangle_{\mathrm{MT}}\right], \tag{25}$$

$$\sigma^{\mathrm{core,corr}}_{\alpha\beta} = -\frac{1}{2\Omega}\sum_{a,c}\int_{R_a} \mathrm{d}^3r\,\rho^a_c(\mathbf{r})(r_\beta\partial_\alpha + r_\alpha\partial_\beta)V_{\mathrm{eff}}(\mathbf{r}). \tag{26}$$

The electrostatic and LDA exchange-correlation pieces are

$$\sigma^{\mathrm{es}}_{\alpha\beta} = -\frac{\delta_{\alpha\beta}}{2\Omega}\int_\Omega \mathrm{d}^3r\,\rho V_C + \frac{1}{2\Omega}\int_\Omega \mathrm{d}^3r\,\rho\,\partial_{\alpha\beta}V_C - \frac{1}{2\Omega}\sum_a Z_a\partial_{\alpha\beta}V_M(\tau_a), \tag{27}$$

$$\sigma_{\alpha\beta}^{\mathrm{xc,LDA}} = \frac{\delta_{\alpha\beta}}{\Omega} \int_{\Omega} \mathrm{d}^3 r \, \rho(\epsilon_{\mathrm{xc}} - \mu_{\mathrm{xc}}). \tag{28}$$

The WIEN2k implementation contains the final explicit forms, including GGA corrections. Those terms should remain in `lapw0`; reimplementing them inside `dmft2` would duplicate density terms and create sign and symmetry inconsistencies.

The APW surface contribution in the current `lapw2` code is a kinetic/boundary term needed by that representation. It should not be confused with the older "discontinuity correction" rejected in the newer Blaha derivation. The new implementation should reproduce the validated Blaha/WIEN2k sum, not revive the older Thonhauser/Klüppelberg electrostatic decomposition.

7 Why the older Klüppelberg electrostatic route is excluded

Klüppelberg's thesis contains a careful and often useful derivation, but its complete stress implementation did not pass its own hydrostatic tests. Blaha and coworkers later attempted to use the same electrostatic formulas and reported huge errors whose precise origin remained unresolved. It would therefore be inaccurate to claim that one printed equation has been proved wrong. What is known is that the failure was localized primarily in the electrostatic/pseudocharge chain and in a large discontinuity contribution, whereas the kinetic and exchange-correlation checks were much better.

The fragile object in the older route can be written schematically as

$$V_C^{\epsilon} = \mathcal{W}_{\epsilon}[\rho_{\epsilon}], \qquad \partial_q V_C = \mathcal{W}_0[\partial_q \rho] + (\partial_q \mathcal{W}_{\epsilon})_0[\rho]. \tag{29}$$

Here $\mathcal{W}_{\epsilon}$ is not just the factor $4\pi/G^2$. It contains the strained muffin-tin/interstitial partition, multipoles, pseudocharge construction, Fourier normalization, boundary matching, Madelung term, and the $G = 0$ convention. Klüppelberg differentiates this construction into many individually large terms whose cancellation is numerically delicate. The tests show that this chain did not produce the correct final pressure, but they do not uniquely identify whether the defect is an algebraic omission, a domain convention, a pseudocharge derivative, or an implementation mismatch.

The validated Blaha representation avoids that cancellation problem. After transforming the strained integrals back to the unstrained cell, it keeps the Fourier coefficients of the real interstitial density and of the auxiliary pseudocharge invariant:

$$\rho_I^{\epsilon}(\mathbf{G}^{\epsilon}) = \rho_I(\mathbf{G}), \qquad \widetilde{\rho}^{\epsilon}(\mathbf{G}^{\epsilon}) = \widetilde{\rho}(\mathbf{G}), \qquad \mathbf{G}^{\epsilon} = F^{-T}\mathbf{G}. \tag{30}$$

Thus the interstitial Coulomb derivative follows only from the reciprocal metric:

$$\boxed{\partial_{\alpha\beta} V_C(\mathbf{G}) = 2\frac{G_\alpha G_\beta}{G^2} V_C(\mathbf{G}) \quad (G \neq 0).} \tag{31}$$

This is the electrostatic baseline already implemented in the modern WIEN2k stress path and should remain owned by `lapw0`.

7.1 Three different "surface" objects must not be mixed

Object	Meaning	Action
APW kinetic surface term	boundary term of the present LAPW kinetic-stress representation; it is on the active `lapw2` call path	retain
Finite-$l_{\max}$ surface/matching terms	genuine corrections associated with a truncated angular representation when required by the chosen kernel	retain exactly where the validated code has them
Older "discontinuity correction"	separate contribution used in the Thonhauser/Klüppelberg formulation; it was numerically huge in failed tests and is absent from the validated Blaha formula	do not add

Klüppelberg remains useful for strain kinematics, spherical-harmonic and radial derivatives, the kinetic stress, the core/Janak relation, and general Pulay bookkeeping. Those parts may be used as derivational checks. The multi-term electrostatic variation and the standalone discontinuity correction are not implementation templates for eDMFT stress.

8 Generalizing the valence terms to eDMFT

8.1 Valence kinetic stress

The kinetic kernel is a one-particle operator, so its eDMFT generalization is immediate:

$$\boxed{\sigma_{\alpha\beta}^{\mathrm{val,kin}} = \frac{1}{2\Omega}\sum_{\mathbf{k},ij}(\rho_{\mathbf{k}})_{ji}\langle\psi_{i\mathbf{k}}|\partial_\alpha\partial_\beta + \partial_\beta\partial_\alpha|\psi_{j\mathbf{k}}\rangle.} \tag{32}$$

The muffin-tin kinetic term, the interstitial kinetic term, and the APW surface term all use ρ only. This makes them the simplest parts to port from `lapw2`.

8.2 Valence IBS/Pulay correction

The DFT expression contains both H and εO. In eDMFT the eigenvalue factor must be replaced by the energy-density matrix:

$$\boxed{\sigma_{\alpha\beta}^{\mathrm{val,corr}} = \frac{2}{\Omega}\,\mathrm{Re}\sum_{\mathbf{k},ij}\left[(\rho_{\mathbf{k}})_{ji}\langle\psi_{i\mathbf{k};\alpha\beta}|H^{\mathrm{KS}}|\psi_{j\mathbf{k}}\rangle_{\mathrm{MT}} - (\rho^{E}_{\mathbf{k}})_{ji}\langle\psi_{i\mathbf{k};\alpha\beta}|\psi_{j\mathbf{k}}\rangle_{\mathrm{MT}}\right].} \tag{33}$$

Here $|\psi_{i;\alpha\beta}\rangle$ is only the explicit strain derivative of the LAPW representation used in the IBS correction. The Blaha derivation shows that the interstitial contribution to this derivative vanishes to first order; the integral is over the muffin-tin region.

It is useful to define two reusable kernels,

$$C^H_{ij,q} = 2\,\mathrm{Re}\langle\psi_{i;q}|H^{\mathrm{KS}}|\psi_j\rangle_{\mathrm{MT}}, \tag{34}$$
$$C^O_{ij,q} = 2\,\mathrm{Re}\langle\psi_{i;q}|\psi_j\rangle_{\mathrm{MT}}, \tag{35}$$

so that

$$\Omega\sigma_q^{\mathrm{val,corr}} = \sum_{\mathbf{k}} \left[\mathrm{Tr}(\rho_{\mathbf{k}} C^H_q) - \mathrm{Tr}(\rho^E_{\mathbf{k}} C^O_q)\right]. \tag{36}$$

Why a naive occupation replacement fails. Replacing `weight(num)` by diagonal elements of `Aweight` would lose off-diagonal coherence and would still multiply the overlap term by the wrong energy. Equation (33) requires the full matrices `Aweight` and `AEweight`.

9 The new explicit projector stress

Let the embedded dynamic potential be

$$\Sigma^{\mathrm{emb}}_{\mathbf{k}}(i\omega_n) = P^\dagger_{\mathbf{k}}\widetilde{\Sigma}(i\omega_n)P_{\mathbf{k}}, \qquad \widetilde{\Sigma} = \Sigma - V_{\mathrm{DC}}. \tag{37}$$

Stationarity removes the derivative of Σ and V_{DC}, but not the explicit geometric derivative of the projector. Hence

$$\partial_q \Sigma^{\mathrm{emb}} = P^\dagger_{;q}\widetilde{\Sigma}P + P^\dagger\widetilde{\Sigma}P_{;q}. \tag{38}$$

The new DMFT stress term is therefore

$$\boxed{\sigma_q^{\mathrm{proj}} = \frac{2}{\Omega\beta}\,\mathrm{Re}\sum_{\mathbf{k},n} e^{i\omega_n 0^+}\,\mathrm{Tr}\left[G_{\mathbf{k}}(i\omega_n)P^\dagger_{\mathbf{k}}\widetilde{\Sigma}(i\omega_n)P_{\mathbf{k};q}\right].} \tag{39}$$

If the implementation explicitly forms both Hermitian-conjugate terms, the factor $2\,\mathrm{Re}$ must be removed. This is a coding convention, not an additional physical factor.

There is no separate first-derivative vertex term. An extra term is needed only if an interaction parameter, Coulomb tensor, or double-counting functional has an explicit strain dependence chosen by the user. For fixed U, J, and a fixed double-counting definition, that term is zero.

10 Derivative of the normalized projector

10.1 Raw projector and global normalization

Write the raw finite-window projector as $P^0_{\mathbf{k}}$ and the projector used by the eDMFT calculation as

$$P_{\mathbf{k}} = P^0_{\mathbf{k}}X, \qquad S = \sum_{\mathbf{k}} w_{\mathbf{k}} P^{0\dagger}_{\mathbf{k}} P^0_{\mathbf{k}}, \qquad X = S^{-1/2}. \tag{40}$$

The **k**-point weights are constants under homogeneous strain in the discrete Brillouin-zone sum:

$$\partial_q w_{\mathbf{k}} = 0. \tag{41}$$

The exact derivative is

$$\boxed{P_{\mathbf{k};q} = P^0_{\mathbf{k};q} X + P^0_{\mathbf{k}} X_{;q}.} \tag{42}$$

The overlap derivative is

$$S_{;q} = \sum_{\mathbf{k}} w_{\mathbf{k}} \left(P^{0\dagger}_{\mathbf{k};q} P^0_{\mathbf{k}} + P^{0\dagger}_{\mathbf{k}} P^0_{\mathbf{k};q} \right). \tag{43}$$

10.2 Matrix inverse-square-root derivative

The derivative of a matrix inverse square root is not an elementwise derivative. If

$$S = V \operatorname{diag}(s_i) V^\dagger, \tag{44}$$

then

$$\left(V^\dagger X_{;q} V \right)_{ij} = f^{[1]}(s_i, s_j) \left(V^\dagger S_{;q} V \right)_{ij}, \qquad f(s) = s^{-1/2}, \tag{45}$$

where

$$f^{[1]}(s_i, s_j) = \begin{cases} \dfrac{s_i^{-1/2} - s_j^{-1/2}}{s_i - s_j}, & s_i \neq s_j, \\ -\dfrac{1}{2} s_i^{-3/2}, & s_i = s_j. \end{cases} \tag{46}$$

An equivalent and numerically useful check follows from $XSX = I$:

$$X_{;q} S X + X S_{;q} X + X S X_{;q} = 0. \tag{47}$$

The present force code returns the relative derivative $B_q = X^{-1} X_{;q}$ through a Sylvester equation. The same routine can be generalized from three displacement components to six strain components.

10.3 Direct, window, and normalization pieces

For implementation it is essential to split

$$P^0_{;q} = P^{0,\mathrm{dir}}_{;q} + P^{0,\mathrm{win}}_{;q}, \tag{48}$$

which gives

$$\boxed{P_{;q} = P^{0,\mathrm{dir}}_{;q} X + P^{0,\mathrm{win}}_{;q} X + P^0 X_{;q}.} \tag{49}$$

The derivative $S_{;q}$ in Eq. (43) must be constructed from the same total raw derivative. In particular, an exact treatment contains a normalization–window cross contribution. One may include it either by explicitly forming $P^{0,\mathrm{win}}_{;q}$ before evaluating $S_{;q}$ or by using the adjoint Fréchet derivative of $S^{-1/2}$ when constructing the outside-window response. It must not be silently omitted.

11 What is contained in the direct projector derivative

For a muffin-tin projector,

$$P^0_{nm} = \langle \psi_{n\mathbf{k}} | \phi_m \rangle_{\mathcal{D}}, \tag{50}$$

where $\mathcal{D}$ denotes the projector domain. Its direct strain derivative contains

$$P^{0,\mathrm{dir}}_{nm;q} = \langle \psi^{\mathrm{geom}}_{n\mathbf{k};q} | \phi_m \rangle_{\mathcal{D}} + \langle \psi_{n\mathbf{k}} | \phi_{m;q} \rangle_{\mathcal{D}} + P^{0,\mathrm{domain}}_{nm;q}. \tag{51}$$

Under the version-1 choices, $\phi_{m;q} = 0$ at fixed atom-centered coordinate and $P^{0,\mathrm{domain}}_{nm;q} = 0$. These two slots are displayed because they make later extensions transparent: $\phi_{m;q}$ contains the moving-local-axis term, whereas $P^{0,\mathrm{domain}}_{nm;q}$ would be needed only for a different moving-cutoff convention.

The first term contains precisely the ingredients appearing in the Blaha LAPW stress derivative:

- derivative of reciprocal vectors $\mathbf{k} + \mathbf{G}$;
- angular derivative of spherical harmonics;
- derivative of spherical Bessel functions and matching coefficients;
- volume-normalization derivative;
- APW and local-orbital contributions;
- derivative of phase factors and symmetry rotations.

The WIEN2k routine `val_corr_stress` already constructs the strain variation of the LAPW matching coefficients, conventionally denoted $D_{lm\lambda}(\alpha, \beta)$. That machinery should be factored into a shared helper and reused when building the projector strain channels.

11.1 Why a finite value at R_{MT} is not a missing term

For a general moving integration domain the Reynolds term would be

$$P^{0,\mathrm{domain}}_{nm;q} = \int_{\partial\mathcal{D}} \mathrm{d}S\, \psi^*_{n\mathbf{k}}(\mathbf{r}) \phi_m(\mathbf{r}) \left(\mathbf{v}_q \cdot \hat{\mathbf{n}}\right). \tag{52}$$

However, the version-1 convention in Eqs. (9)–(12) evaluates the projector in the current atom-centered coordinate over the fixed ball $|\mathbf{x}| < R_{\mathrm{MT}}$. In that coordinate the normal boundary velocity is zero and therefore

$$P^{0,\mathrm{domain}}_{nm;q} = 0 \qquad \text{(fixed current sphere)}. \tag{53}$$

The nonzero projector value at the sphere edge is consequently harmless for this particular derivative. It would matter if the cutoff radius were rescaled, if a material sphere were pulled forward by F, or if the inside and outside representations were differentiated inconsistently. For projectors 5/6 the tail must be included, so R_{MT} is not a support boundary at all.

The analytic result must still be checked by rebuilding the raw projector for the $\pm\delta\epsilon_q$ structures with exactly the same radial file and radii. That finite difference tests all reciprocal-vector, phase, matching, angular, and tail terms without adding a spurious transport contribution.

12 Local axes and a diagonal-only self-energy

This issue should be separated cleanly from the base stress derivation. Suppose Q is a projector expressed in a fixed global orbital basis and $D(\epsilon)$ is the unitary orbital representation of a geometry-following local frame. The projector in that moving local frame is

$$P(\epsilon) = D^\dagger(\epsilon) Q(\epsilon). \tag{54}$$

Define the anti-Hermitian connection

$$A_q = D^\dagger D_{;q}, \qquad A_q^\dagger = -A_q. \tag{55}$$

Then the projector derivative separates into

$$\boxed{P_{;q} = D^\dagger Q_{;q} - A_q P.} \tag{56}$$

The first term is the fixed-axis derivative implemented in version 1. The second is the later axis-motion correction. If the local dynamic potential is diagonal,

$$\widetilde{\Sigma}_{mm'}(i\omega_n) = \delta_{mm'} \widetilde{\Sigma}_m(i\omega_n), \tag{57}$$

then the same correction can be viewed in the fixed global basis as

$$\left(D\widetilde{\Sigma}D^\dagger\right)^{\mathrm{axis}}_{;q} = D[A_q, \widetilde{\Sigma}]D^\dagger, \tag{58}$$

with

$$[A_q, \widetilde{\Sigma}]_{mm'} = (A_q)_{mm'} \left(\widetilde{\Sigma}_{m'} - \widetilde{\Sigma}_m\right). \tag{59}$$

Thus the correction vanishes for an orbitally isotropic self-energy but is generally nonzero for an anisotropic diagonal self-energy under a shear that rotates the local ligand frame. Substitution of $P^{\mathrm{axis}}_{;q} = -A_q P$ into Eq. (39) is sufficient; it should be added to the direct projector derivative rather than coded as a second unrelated stress formula. Equivalently, the explicit embedded-potential derivative is

$$\left(\Sigma^{\mathrm{emb}}\right)^{\mathrm{axis}}_{;q} = P^\dagger [A_q, \widetilde{\Sigma}] P. \tag{60}$$

12.1 How to define a smooth geometry-following frame

A moving frame must be a deterministic, differentiable function of atomic geometry. For a normalized ligand direction $\mathbf{u} = \mathbf{d}/|\mathbf{d}|$ that follows the homogeneous strain,

$$\partial_q \mathbf{u} = \left(I - \mathbf{u}\mathbf{u}^T\right) E_q \mathbf{u}, \tag{61}$$

where E_q is the symmetric generator associated with channel q. Two or more such directions can be assembled into a raw frame B and orthogonalized smoothly by the polar construction

$$R = B(B^T B)^{-1/2}. \tag{62}$$

Its derivative uses the same Fréchet/Sylvester machinery as the projector normalization. The Cartesian angular velocity is

$$\Omega_q = R^T R_{;q}, \qquad \Omega_q^T = -\Omega_q, \tag{63}$$

and is converted to the orbital connection A_q through the l-shell rotation generators (or by differentiating the same Wigner matrix used by the projector code). Signs must be verified against Eq. (56) by finite differences.

Do not define the moving frame by independently diagonalizing a nearly degenerate crystal-field matrix at every strain. Eigenvectors can swap, change sign, or rotate arbitrarily inside a degenerate subspace, making the numerical derivative discontinuous. A ligand-based smooth frame or a parallel-transported frame is preferable.

12.2 Recommended hierarchy

1. **Version 1:** fixed Cartesian local axes, $A_q = 0$, diagonal/block self-energy allowed.
2. **Version 2:** optional geometry-following axes with Eq. (56), tested first for a prescribed analytic rotation.
3. **Long-term covariant solution:** full local self-energy matrix and rotationally invariant interaction. Then a complete-shell frame rotation is a gauge transformation and the physical stress is frame independent, up to subspace truncation and numerical errors.

13 Fixed band-index window and outside-band response

The DMFT window is a fixed band-index interval, for example bands 65–164. Window membership therefore does not have an energy-cutoff derivative. The wave functions inside the fixed window nevertheless mix with bands outside it.

For $n \in W_{\mathbf{k}}$,

$$\boxed{|\psi_{n\mathbf{k};q}\rangle_Q = \sum_{a\notin W_{\mathbf{k}}}^{\text{stored}} |\psi_{a\mathbf{k}}\rangle \frac{\langle\psi_{a\mathbf{k}}|H_{\mathbf{k};q} - \varepsilon_{n\mathbf{k}} O_{\mathbf{k};q}|\psi_{n\mathbf{k}}\rangle}{\varepsilon_{n\mathbf{k}} - \varepsilon_{a\mathbf{k}}}.} \tag{64}$$

The sum need not use an iterative Sternheimer solver. It can use every extra band already written by `lapw1` to `case.vector` or `case.vectorso`. Convergence is improved by asking `lapw1` to write more bands while leaving the DMFT window itself unchanged.

If $U_{am} = \langle \psi_a | \phi_m \rangle$, Eq. (64) implies

$$(P^{0,\mathrm{win}}_{;q})_{nm} = \sum_{a \notin W} C^{q*}_{an} U_{am}, \qquad C^q_{an} = \frac{H^q_{an} - \varepsilon_n O^q_{an}}{\varepsilon_n - \varepsilon_a}. \tag{65}$$

The current `BuildWindowResponseMatrices` implements an adjoint version of this correction. It builds two Hermitian matrices ρ^{resp} and e^{resp} such that the response can be evaluated with the same geometric kernels:

$$\Delta^{\mathrm{window}}_q = \mathrm{Tr}\left[\rho^{\mathrm{resp}} H_{;q} - e^{\mathrm{resp}} O_{;q}\right]. \tag{66}$$

This routine is perturbation independent and is therefore directly reusable for stress. Its output must be passed through stress-only contractions and must not be added to the physical density.

Cross-window degeneracy. If $\varepsilon_n \simeq \varepsilon_a$ at the window boundary, simply skipping the denominator is not exact. The final production version should detect a degenerate cluster and perform degenerate perturbation theory in the combined inside/outside subspace. The present force code reports and skips denominators below a threshold; that is acceptable only as a diagnostic first step.

14 A practical contraction strategy

There are two possible implementations of Eq. (21).

14.1 Option A: construct full derivative matrices

Build $H_{;q}$ and $O_{;q}$ in the stored band basis and evaluate traces directly. This is algebraically transparent and also provides the coefficients C^q_{an} needed by Eq. (65). Its disadvantages are the N_b^2 storage and the need to generalize all LAPW stress kernels to mixed bra/ket band pairs.

14.2 Option B: pseudo-band diagonalization

Any Hermitian matrix can be diagonalized,

$$\rho = V_\rho \,\mathrm{diag}(r_p) V_\rho^\dagger, \qquad \rho^E = V_E \,\mathrm{diag}(e_p) V_E^\dagger. \tag{67}$$

Transform the LAPW coefficients by V_ρ and evaluate H-like kernels on the resulting pseudo-states with signed weights r_p. Transform a second copy by V_E and evaluate overlap-like kernels with signed weights e_p. Then

$$\mathrm{Tr}(\rho C^H_q) - \mathrm{Tr}(\rho^E C^O_q) = \sum_p r_p \,\widetilde{C}^H_{pp,q} - \sum_p e_p \,\widetilde{C}^O_{pp,q}. \tag{68}$$

The same procedure works for the indefinite Hermitian matrices ρ^{resp} and e^{resp}. Negative eigenvalues are not a problem; they are response weights, not probabilities.

Recommended first implementation. Use Option B for the physical and window-response stress. It reuses the existing `Diagonalize_DMFT_WEIGHTS` and `cmp_alm_blm_clm` machinery and avoids storing full stress matrices. Split `val_corr_stress` into energy-independent H and overlap pieces. Reserve Option A, or a small outside–inside mixed matrix, for exact normalization–window and degeneracy handling if the adjoint implementation is not extended.

15 What WIEN2k `lapw2` already provides

The relevant source is `~/wien2k_24/SRC_lapw2`. The existing path recognizes modes `STR` and `STRF`, allocates the stress buffers, prepares radial integrals and plane-wave derivatives, evaluates four valence pieces, symmetrizes, divides by the volume, and writes `:STR_VAL001-003`.

WIEN2k routine/module	What it supplies	eDMFT action
`stress_kb`, `stress_pw`	radial stress integrals; $\mathbf{k}+\mathbf{G}$, first and second Bessel derivatives	import modules and allocation/finalization logic
`harmon.f`/`DDVBES1`	$F1P$, `DFJ_ST`, `DDFJ_ST`, second spherical-Bessel derivatives	adapt `dmft2/harmon.f90`; keep force path unchanged
`val_corr_integrals`	radial integrals needed by the IBS correction	copy/adapt essentially unchanged
`kin_str_integrals`	radial integrals for muffin-tin kinetic stress	copy/adapt essentially unchanged
`val_kinstrmt`	muffin-tin kinetic tensor per state	reuse with ρ pseudo-states
`val_kin_interstitial`	interstitial kinetic tensor using the step function	reuse with ρ pseudo-states; audit indices and scaling
`stress_APW_surface`	APW kinetic surface/boundary contribution	reuse with ρ pseudo-states; reproduce DFT output first
`val_corr_stress`	basis/matching derivative and DFT $(H-\varepsilon O)$ IBS correction	refactor into H and O kernels; also expose matching derivatives for projector stress
`c_alpha_m` and angular helpers	Cartesian/spherical angular algebra	copy/adapt unchanged
`symmetry_stress_rotij`	rotation of equivalent-atom tensors	reuse or carefully simplify
`symmetrization_stress`	point-group average, volume division, `STR_VAL` output	reuse behavior and output format; clean internal API
`l2main.F` stress blocks	call order, signs, factors of two, and accumulation	use as the DFT reference implementation

The Makefile also lists older FFT stress helpers. Only routines on the actual call path should be imported. In particular, current `val_kin_interstitial` calls `step_fun` directly; unused legacy helpers should not be copied merely because they are listed.

15.1 A source-level caution about APW surface data

The `A_surf2` data are built in the typed `l2main.F` path via the `!_REAL`/`!_COMPLEX` extraction rule in the WIEN2k Makefile. Lines that look commented in the generic source become active in the selected

real or complex build. The port must preserve this behavior explicitly rather than copying the apparent comments verbatim.

15.2 A source-level caution about validation

The stress routines are old Fortran and contain historical/dead test code. The goal is not blind textual copying. The first gate must be exact reproduction of the current WIEN2k DFT `:STR_VAL` for the same vectors, occupations, and structure. Only after that gate is passed should the eDMFT contraction be enabled.

16 What is already available in `dmft2`

The relevant source is `~/projects/eDMFT_new/src/dmft2`.

Existing object/routine	Why it matters for stress	Required change
`Aweight`	full Hermitian interacting density matrix, including off-diagonal band coherence and $\mathbf{k}$ weight	compute for stress as now for force
`AEweight`	full Hermitian energy-weighted density matrix	enable under `Qstress`; preserve tail convention
`Diagonalize_DMFT_WEIGHTS`	converts a Hermitian band matrix into pseudo-state weights and eigenvectors	reuse for ρ, ρ^E, and response matrices
`cmp_alm_blm_clm`	transforms LAPW $A/B/C$ coefficients by an arbitrary band rotation	reuse for the two pseudo-band contractions
`Force1_DMFT`	demonstrates the correct $\rho H - \rho^E O$ off-diagonal contraction in the force path	use as design reference for stress
`Build_DMFT_Projector`	builds raw projector and three force derivative channels	generalize or add a separate six-channel strain builder
`RenormalizeTrans` / `RenormalizeTransK`	applies `SOlapm.dat` or SVD-like normalization to projectors	reuse, but differentiate normalization correctly
`ComputeNormalization Derivative`	Sylvester/Fréchet derivative of $S^{-1/2}$ for three force directions	generalize to an arbitrary number of derivative channels
`AddProjector NormalizationDerivative`	adds the missing normalized-projector derivative to force channels	add a strain analogue with the correct channel convention
`BuildWindowResponse Matrices`	uses extra vector-file bands to avoid an iterative Sternheimer solve	reuse for six stress contractions; extend exact normalization coupling
`DMFTUfull`	projectors for all stored vector bands, not only the DMFT window	reuse for outside-window sums

Existing object/routine	Why it matters for stress	Required change
`cmpLogGdloc` and `Gdloc(:,:,ip)`	evaluates explicit projector derivative channels with controlled frequency tails	generalize hard-coded four-channel assumptions
`Reduce_MPI`	combines distributed **k**-point results	extend to stress tensors and projector-stress accumulators
`w_RHOLM`, `sumfft`	converged eDMFT valence density sent to the rest of the WIEN2k cycle	no new stress formula; ensure `lapw0` sees this density

Several present assumptions are force specific:

- `nipc=4` means base plus x, y, z;
- `CompressSigmaTransformation2` executes derivative algebra only when `nipc.eq.4`;
- force channels use the convention $U_A = -iD_A$;
- `cmpLogGdloc` contains arrays dimensioned `0:3` and labels `c,x,y,z`;
- `AEweight` construction is currently guarded by `Qforce`;
- the mode and input paths do not yet recognize `STR`/`STRF`.

All of these must be generalized before stress can be correct.

17 Recommended derivative-channel design

A minimal extension would set

$$\texttt{nipc} = 1 + 3 + 6 = 10 \tag{69}$$

for a simultaneous force and stress run. This works, but it makes every consumer infer meaning from a channel number and unnecessarily enlarges frequency-dependent arrays.

The clearer design is to keep separate objects:

```
P0 ! normalized ordinary projector
dP_force(:,:,:,3)
dP_strain(:,:,:,6)
lgTrans_force(:,:,:,3)
lgTrans_strain(:,:,:,6)
```

with metadata

```
strain_alpha = [1,2,3,2,1,1]
strain_beta = [1,2,3,3,3,2]
```

If a unified fourth dimension is retained for minimal disruption, add explicit offsets and channel kind arrays. Never use `nipc.eq.4` as a physical test.

For a strain channel stored as the actual derivative $P_{;q}$, the local derivative contraction is

$$L_q = P^\dagger_{;q} P + P^\dagger P_{;q}. \tag{70}$$

The force code instead stores D_A and defines $P_{;A} = -iD_A$. This explains its present difference-like combination multiplied by i. Stress code must not reuse that algebra unless it also adopts the same artificial convention. The simplest choice is to store actual strain derivatives and use Eq. (70).

18 Detailed implementation plan

18.1 Stage 0: freeze conventions and create diagnostics

Before changing physics code, document and print:

- six-component strain map and shear convention;
- stress sign and pressure sign;
- whether ρ and ρ^E already contain the $\mathbf{k}$ weight;
- assert the version-1 choices: $\partial_q R_{\mathrm{MT}} = 0$, current spherical domains, fixed `projectorw.dat`/R_2, and fixed Cartesian local axes;
- whether a run is `STR`, `FOR`, or `STRF`;
- individual tensors before point-group symmetrization.

No new DMFT formula should be activated in this stage.

18.2 Stage 1: reproduce DFT valence stress inside `dmft2`

1. Add `Qstress` and recognize `STR`/`STRF`.
2. Import/adapt the preparation and angular modules `stress_kb`, `stress_pw`, and `c_alpha_m`.
3. Import/adapt the radial-integral and kinetic kernels `val_corr_integrals`, `kin_str_integrals`, `val_kinstrmt`, `val_kin_interstitial`, and `stress_APW_surface`.
4. Import/adapt `val_corr_stress` and the stress symmetrization helpers.
5. Extend `harmon.f90` with the second Bessel derivatives and stress plane-wave arrays.
6. Add the two radial-integral calls to the `dmft2` atomic setup, mirroring `lapw2/atpar.F`.
7. Use diagonal DFT occupations and eigenvalues, not eDMFT matrices.
8. Write `:STR_VAL001-003` in exactly the WIEN2k format.

Gate 1: With self-energy disabled, every individual valence piece and their sum must agree with ordinary `lapw2` to numerical tolerance.

18.3 Stage 2: replace DFT occupations by ρ for kinetic pieces

1. Ensure `Aweight` is constructed when `Qstress` is true.
2. Diagonalize `Aweight`.
3. Transform plane-wave and muffin-tin coefficients into pseudo-states.
4. Evaluate muffin-tin kinetic, interstitial kinetic, and APW surface kernels with the signed eigenweights.
5. Keep the normal density construction unchanged.

Gate 2: In the diagonal DFT limit this path must reduce exactly to Stage 1. Under a nonzero self-energy, the tensor must be Hermitian/symmetric after the standard symmetry operations.

18.4 Stage 3: generalize the valence correction

Refactor `val_corr_stress` so that every occurrence of the band eigenvalue is separated from the energy-independent part. Conceptually return

$$C_q^{\mathrm{DFT}} = C_q^H - \varepsilon C_q^O. \tag{71}$$

Then:

1. evaluate C^H using pseudo-states from `Aweight`;
2. evaluate C^O using pseudo-states from `AEweight`;
3. combine them with the minus sign in Eq. (36);
4. include completely occupied bands below the DMFT window as diagonal DFT additions, exactly as in the force path.

Gate 3: Set $\rho_{ij} = f_i\delta_{ij}$ and $\rho^E_{ij} = f_i\varepsilon_i\delta_{ij}$. The refactored result must reproduce the original unsplit `val_corr_stress`.

18.5 Stage 4: add finite-window response to the valence stress

1. Call `BuildWindowResponseMatrices` whenever stress is active and more vector bands than DMFT bands are available.
2. Diagonalize ρ^{resp} and use the stress-only H kernels.
3. Diagonalize e^{resp} and use the stress-only overlap kernels.
4. Do not accumulate charge, local density, energy, or `RHOLM` from these response pseudo-states.
5. Print the response norm, number of outside bands, smallest denominator, and number of skipped/degenerate pairs.

Gate 4: Increase the number of bands written by `lapw1` while keeping the DMFT window fixed. The stress correction should converge smoothly.

18.6 Stage 5: build six direct projector strain channels

1. Factor the Blaha matching-coefficient derivative out of `val_corr_stress` into a reusable helper.
2. Extend `Build_DMFT_Projector`, or add `Build_DMFT_Projector_Strain`, to construct $P^{0,\mathrm{dir}}_{;q}$ for all six q in one pass.
3. Include the APW/local-orbital, projector-5/6 tail, phase, and reciprocal-vector pieces.
4. Include the angular, Bessel-argument, matching, and volume-normalization pieces according to Stage 0. Do not differentiate the stored radial projector or add a moving-boundary term.
5. Normalize the base projector with the saved `SOlapm.dat`; do not silently reconstruct a different projector during a stress-only step.

Gate 5: At each test **k** point compare analytic raw-projector derivatives with central finite differences of a projector rebuilt at $\pm\delta\epsilon_q$ using identical radial files, R_{MT}, and R_2. Test P^0, S, X, and the final normalized P separately.

18.7 Stage 6: add the normalization derivative

1. Generalize `ComputeNormalizationDerivative` from a trailing dimension 3 to an arbitrary number of derivative channels.
2. Form $S_{;q}$ from the raw strain channels.
3. Solve the same Sylvester/Fréchet problem used by the corrected force implementation.
4. Add $P^0 X_{;q}$ to every normalized strain channel.
5. Verify Eq. (47) numerically and print its residual.

Gate 6: The finite-difference derivative of $P^\dagger P$ and the analytic derivative must agree. With a complete window and identity normalization, this correction must vanish.

18.8 Stage 7: close the normalization–window coupling

There are two exact routes:

Explicit route
Build the outside–inside $H^q_{an} - \varepsilon_n O^q_{an}$ elements, form $P^{0,\mathrm{win}}_{;q}$ using Eq. (65), include it in $S_{;q}$, and add the resulting $X^{\mathrm{win}}_{;q}$.

Adjoint route
Differentiate the normalization map in reverse mode. Apply the adjoint Fréchet derivative of $S^{-1/2}$ to the projector gradient before building the perturbation-independent response matrices. This folds the normalization–window term into `BuildWindowResponseMatrices` without forming six explicit outside-band derivatives.

The adjoint route is likely faster and fits the present response architecture; the explicit route is easier to verify. A good development sequence is to implement the explicit route for small test cases, validate it, and then use it as a reference for the adjoint optimization.

18.9 Stage 8: evaluate the projector stress with correct tails

1. Generalize `CompressSigmaTransformation2` to accept actual strain derivatives and Eq. (70).
2. Remove hard-coded `nipc.eq.4` branches and labels.
3. Generalize `Gdloc`, `Nds`, and `cmpLogGdloc` derivative dimensions or provide a dedicated stress channel object.
4. Reuse the force code's analytic high-frequency treatment of the logarithmic term.
5. Do not divide the result elementwise by diagonal entries of `SOlapm`. The channels already represent the normalized projector.

Gate 8: Compare Eq. (39) with a central finite difference of only the embedded self-energy/projector part while G, Σ, and V_{DC} are held fixed.

18.10 Stage 8b: optional geometry-following local axes

This stage is deliberately postponed until the fixed-axis stress passes all preceding gates.

1. Define a smooth geometry-to-frame rule, preferably from ligand vectors and polar orthogonalization as in Eqs. (61)–(62).
2. Compute the six anti-Hermitian connections $A_q = D^\dagger D_{;q}$.
3. Add $-A_q P$ to the six direct normalized-projector channels before the logarithmic/projector contraction.
4. For debugging, independently evaluate $P^\dagger[A_q, \widetilde{\Sigma}]P$ and verify equality with the two projector derivative terms.
5. Confirm that the correction vanishes for an orbital-independent $\widetilde{\Sigma}$ and transforms covariantly under a constant change of local basis.

Gate 8b: Prescribe a small analytic frame rotation without changing the lattice and compare the commutator formula, the projector-channel formula, and a central finite difference. Only then test a ligand-following frame under physical shear.

18.11 Stage 9: reduce, symmetrize, write, and assemble

1. Extend MPI reduction to every unsymmetrized tensor piece.
2. Rotate equivalent-atom contributions exactly as `lapw2` does.
3. Apply the crystal point-group average.
4. Add the projector tensor to the generalized valence tensor.
5. Divide by Ω exactly once.
6. Write `:STR_VAL001-003`.
7. Run normal `lcore` and `lapw0` stress steps and let the existing mixer assemble the final tensor.

The current mixer convention is

$$\texttt{Tstress} = \texttt{tot_val_st} + \texttt{cor_corr} - \texttt{ES_STRESS}. \tag{72}$$

Therefore `dmft2` should match the sign and units of ordinary `lapw2 :STR_VAL`; it should not attempt to output the final total stress itself.

19 Proposed routine-level changes

This table is a design map, not a request to edit the files immediately.

File/routine	Planned responsibility
`l2main.f90`	add `Qstress`; allocate tensors; request `AEweight`; orchestrate physical, window, and projector stress; reduce and write output
`modules.f90`	add/adapt `stress_kb` and `stress_pw`; define six-channel metadata and stress accumulators
`harmon.f90`	compute $\mathbf{k}+\mathbf{G}$, first and second Bessel derivatives needed by stress, guarded by `Qstress`
`atpar` path	call `val_corr_integrals` and `kin_str_integrals` once per atom
new/adapted stress source files	port `val_kinstrmt`, `val_kin_interstitial`, `stress_APW_surface`, angular helpers, and symmetrization
`val_corr_stress` adaptation	expose H and O kernels separately; factor out matching-coefficient derivatives for projector channels
`cmp_dmft_weights.f90`	replace `if(Qforce)` guards by a derivative-moment condition covering force or stress
`l2mainSubs.f90`: projector builder	add six raw strain derivatives and explicit channel semantics
local-frame helper (later stage)	construct a smooth geometry-following frame, its six connections A_q, and add $-A_qP$ to the direct projector channels; inactive for version 1
`ComputeNormalization Derivative`	accept n_q instead of a fixed dimension 3; add residual diagnostics
`BuildWindowResponse Matrices`	reuse for stress; later extend normalization-adjoint and degenerate-boundary handling
`CompressSigma Transformation2`	separate base, force, and strain algebra; remove `nipc.eq.4` meaning
`cmp_log_gdloc.f90`	dynamic derivative-channel dimensions, stress labels, and output accumulators; preserve high-frequency normalization
`Makefile`	add only the stress objects actually called in the new path

20 Terms that should not be added to `dmft2`

The following would double count existing WIEN2k functionality:

- a second electrostatic stress evaluated from `RHOLM`;
- a second LDA/GGA XC stress;
- a second core correction;
- a separate final pressure/sign conversion;
- a derivative of the converged impurity self-energy using a vertex;
- an occupation-number derivative outside the stationary finite-temperature functional;
- the older discontinuity correction absent from the newer Blaha formula.

The only caveat is workflow: `lapw0` and `lcore` must be run in stress mode on the same converged eDMFT density and structure before the mixer assembles the final result.

21 Computational cost

The stress calculation is a post-processing step after convergence. It should not approach the cost of solving the impurity problem.

- The expensive Matsubara Green-function calculation already produces `Aweight` and, when requested, `AEweight`.
- Two or four additional Hermitian diagonalizations of band matrices per $\mathbf{k}$ point scale as N_W^3 and are normally modest.
- WIEN2k computes all nine displayed tensor components in one pass; six strain components do not require six independent `dmft2` runs.
- The interstitial kinetic term and step-function convolution may be a significant fraction of the new time; it should be timed separately.
- The finite-window correction uses bands already stored in the vector files and avoids an iterative Sternheimer solve.
- Six projector derivative channels increase projector-building work and memory, but the local-orbital dimension is small.

A realistic expectation is an order-unity increase in the final `dmft2` post-processing time, not an order-of-magnitude increase in the total converged DFT+eDMFT calculation. The exact factor will be controlled by the number of stored vector bands and the interstitial stress kernel.

22 Validation hierarchy

The following tests should be passed in order. A later test is not a substitute for an earlier one.

22.1 Level 1: algebra and channel identities

- Hermiticity of `Aweight`, `AEweight`, response matrices, S, and $S_{;q}$.
- Residual of Eq. (47).

- Equality between explicit two-term projector derivatives and the $2\,\mathrm{Re}$ form.
- Exact six-to-nine component mapping and shear factors.
- Version-1 assertions: $\partial_q R_{\mathrm{MT}} = \partial_q R_2 = 0$ and $A_q = 0$.

22.2 Level 2: DFT reduction

With $\Sigma - V_{\mathrm{DC}} = 0$ and diagonal DFT weights:

- projector stress must vanish;
- generalized kinetic pieces must equal ordinary `lapw2`;
- the split H/O valence correction must equal the original routine;
- total `:STR_VAL` must match ordinary `lapw2`.

22.3 Level 3: projector finite differences

For each of the six strain modes compare central finite differences with analytic derivatives of

$$P^0, \qquad S, \qquad X = S^{-1/2}, \qquad P = P^0 X, \tag{73}$$

using several step sizes, for example 10^{-3}, 5×10^{-4}, 2×10^{-4}, and 10^{-4}. Separate direct, boundary, window, and normalization pieces in the diagnostic output. For the adopted current-sphere convention the independently reported boundary piece must be zero. The strained calculations must reuse the identical `projectorw.dat`, R_{MT}, R_2, band indices, and local axes.

22.4 Level 4: stationary-functional finite differences

After full electronic convergence at each strained structure, compare the analytic stress with central finite differences of the same stationary finite-temperature eDMFT functional. Do not compare to a nonstationary energy formula with inconsistent entropy or chemical-potential conventions.

For metals, use the same finite-temperature/free-energy functional and smearing in both the analytic and finite-difference calculations.

22.5 Level 5: physical and symmetry tests

- Cubic system: equal diagonal components and zero shear.
- Low-symmetry system: $\sigma_{\alpha\beta} = \sigma_{\beta\alpha}$.
- Hydrostatic strain: $P = -\operatorname{Tr}\sigma/3$ agrees with the equation of state.
- Volume-conserving tetragonal and shear strains: individual components agree with finite differences.
- Basis convergence: stress converges with $R_{\mathrm{MT}}K_{\max}$, $l_{\max}$, radial mesh, $\mathbf{k}$ mesh, Matsubara tail, and stored bands.
- Projector convergence: repeat with/without normalization, enlarged DMFT window, and enlarged stored-band set.

- Diagonal self-energy: compare fixed-axis and optional moving-axis results; the difference must equal the commutator contribution in Eq. (60).

22.6 Suggested material sequence

1. an uncorrelated high-symmetry reference that ordinary WIEN2k handles well;
2. cubic $SrVO_3$ with one correlated atom and vanishing shear by symmetry;
3. a tetragonally strained correlated material;
4. a low-symmetry cell with correlated and uncorrelated atoms and nontrivial projector normalization;
5. spin–orbit cases only after the scalar-relativistic path is validated.

23 Settled choices and deliberately deferred extensions

The geometric conventions are no longer open for the first implementation:

- R_{MT} and the projector-5/6 outer radius are fixed in bohr;
- the integration domains are current atom-centered spheres;
- the stored radial projector is fixed and is not regenerated under strain;
- the DMFT window is a fixed band-index interval;
- local correlated axes are fixed in global Cartesian space;
- the target is the same stationary finite-temperature functional used by the converged eDMFT calculation.

The following are real later extensions, not prerequisites for version 1:

1. **Geometry-following local axes.** Add the smooth-frame connection A_q and the projector correction $-A_q P$, especially for an anisotropic diagonal-only self-energy.
2. **Degenerate window boundary.** Replace denominator skipping by cluster perturbation theory, or enlarge the fixed band interval so that the near-degenerate cluster lies wholly inside it.
3. **Spin–orbit path.** Validate the scalar-relativistic implementation first, then repeat every projector and symmetry test in the spinor representation.
4. **Potential-dependent radial projectors.** If one later chooses to regenerate `projectorw.dat` under strain, derive and implement the radial response and its normalization consistently.
5. **Strain-dependent interactions.** If U, J, the Coulomb tensor, or double counting is explicitly made geometry dependent, add the corresponding explicit derivative of the local functional.

24 Compact formula sheet

$$\sigma_q = \Omega^{-1}\partial_q \Gamma, \tag{74}$$

$$R_{\mathrm{MT};q} = R_{2;q} = 0, \qquad \mathcal{D}_a^{\epsilon} : \ |\mathbf{r}^{\epsilon} - \tau_a^{\epsilon}| < R_a, \tag{75}$$

$$\Omega\sigma_q^{\mathrm{one}} = \sum_{\mathbf{k}} \mathrm{Tr}(\rho_{\mathbf{k}} H_{\mathbf{k};q} - \rho^E_{\mathbf{k}} O_{\mathbf{k};q}), \tag{76}$$

$$\sigma_q^{\mathrm{proj}} = \frac{2}{\Omega\beta}\,\mathrm{Re}\sum_{\mathbf{k},n} \mathrm{Tr}[G_{\mathbf{k}} P^{\dagger}_{\mathbf{k}} \widetilde{\Sigma} P_{\mathbf{k};q}], \tag{77}$$

$$P_{\mathbf{k}} = P^0_{\mathbf{k}} X, \qquad S = \sum_{\mathbf{k}} w_{\mathbf{k}} P^{0\dagger}_{\mathbf{k}} P^0_{\mathbf{k}}, \qquad X = S^{-1/2}, \tag{78}$$

$$P_{\mathbf{k};q} = P^{0,\mathrm{dir}}_{\mathbf{k};q} X + P^{0,\mathrm{win}}_{\mathbf{k};q} X + P^0_{\mathbf{k}} X_{;q}, \tag{79}$$

$$S_{;q} = \sum_{\mathbf{k}} w_{\mathbf{k}} (P^{0\dagger}_{\mathbf{k};q} P^0_{\mathbf{k}} + P^{0\dagger}_{\mathbf{k}} P^0_{\mathbf{k};q}), \tag{80}$$

$$(V^{\dagger} X_{;q} V)_{ij} = f^{[1]}(s_i, s_j)(V^{\dagger} S_{;q} V)_{ij}, \qquad f(s) = s^{-1/2}, \tag{81}$$

$$|\psi_{n;q}\rangle_Q = \sum_{a \notin W} |\psi_a\rangle \frac{\langle \psi_a | H_{;q} - \varepsilon_n O_{;q} | \psi_n \rangle}{\varepsilon_n - \varepsilon_a}, \tag{82}$$

$$\Omega\sigma_q^{\mathrm{val,corr}} = \mathrm{Tr}(\rho C^H_q) - \mathrm{Tr}(\rho^E C^O_q), \tag{83}$$

$$\Delta_q^{\mathrm{window}} = \mathrm{Tr}(\rho^{\mathrm{resp}} H_{;q} - e^{\mathrm{resp}} O_{;q}), \tag{84}$$

$$P^{\mathrm{axis}}_{;q} = -A_q P, \qquad A_q = D^{\dagger} D_{;q}, \qquad (\Sigma^{\mathrm{emb}})^{\mathrm{axis}}_{;q} = P^{\dagger}[A_q, \widetilde{\Sigma}] P. \tag{85}$$

25 Definition of done

The implementation is complete only when all of the following are true:

- `dmft2` reproduces ordinary WIEN2k `:STR_VAL` in the DFT limit;
- ρ and ρ^E are both used, with full off-diagonal coherence;
- the normalized projector derivative contains direct, finite-window, and normalization pieces, including their cross coupling;
- the projector derivative is independently verified by finite differences;
- those finite differences keep R_{MT}, R_2, the radial projector, band indices, and version-1 local axes fixed;
- the logarithmic projector term uses the normalized channel without naive diagonal `SOlapm` division;
- `lapw0`, `lcore`, and the mixer supply the remaining density/core terms exactly once;
- the analytic tensor agrees with finite differences of the stationary eDMFT functional for hydrostatic, tetragonal, and shear strains;
- the result converges with both LAPW basis size and the number of stored outside bands.

The optional moving-axis extension is complete only when its projector derivative and commutator forms agree, its finite difference is smooth through the chosen geometry range, and the correction vanishes for an isotropic local self-energy.

26 Primary references and inspected implementation

1. K. Belbase, A. Tröster, and P. Blaha, "Stress tensor in the linearized augmented plane wave method," *Phys. Rev. B* 104, 174113 (2021). Primary DFT stress derivation used here.

2. T. Thonhauser, C. Ambrosch-Draxl, and D. J. Singh, "Stress and pressure within the linearized augmented plane wave method," *Solid State Communications* 124, 275–282 (2002). Useful historical derivation; the newer Blaha electrostatic treatment takes precedence.

3. A. Klüppelberg, *Calculation of stress tensor within the ab initio total-energy and force code FLEUR*, diploma thesis (2012). Useful for kinematics, kinetic/core/Pulay derivations, and for the documented failed numerical tests; its multi-term electrostatic variation and standalone discontinuity correction are not used as the baseline.

4. A. Klüppelberg's later thesis on forces, phonons, and density-functional perturbation theory. Useful for finite-$l_{\max}$ and response ideas, but not an independent validation of the failed 2012 electrostatic stress implementation.

5. `Forces_LAPW.pdf`, the stationary DFT+eDMFT force derivation and implementation notes.

6. The DFT+eDMFT projector-Pulay force-correction note and the current `eDMFT_new/src/dmft2` source, for finite-window and normalization corrections.

7. `wien2k_24/SRC_lapw2`, `SRC_lapw0`, `SRC_lcore`, and `SRC_mixer`, inspected as the source reference for the existing DFT stress path and output convention.

8. The `dmft2` files `p_project.f90`, `l2mainSubs.f90`, and `dmfts.f90`, inspected for the projector-5/6 radial tail and the present local-axis convention.

This document is an implementation plan. It does not modify source code. Version-1 geometry conventions are fixed here; implementation should begin only after the staged DFT-baseline tests and diagnostics are agreed upon.